

Companion XV

The Early-Universe Gravitational-Wave Spectrum in the Relaxation-Driven Cyclic Cosmology

Relaxation-Induced Damping, Sectoral Asymmetry, and Observable Signatures

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Abstract

The early-Universe gravitational-wave (GW) spectrum provides a unique window into the microphysical processes that shaped the primordial plasma. In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduced in Companion X modifies the evolution of tensor perturbations during the earliest stages of cosmic history. The relaxation rate $\Gamma(a)$ introduces an additional damping term in the tensor-mode equation of motion, suppressing the amplitude of primordial gravitational waves on scales that re-enter the horizon during the relaxation-dominated epoch. Combined with the sectoral temperature asymmetry between the two CPT-conjugate sectors, this leads to a characteristic sectoral GW asymmetry and a relaxation-imprinted suppression feature in the stochastic gravitational-wave background (SGWB).

In this Companion we derive the relaxation-corrected tensor-mode evolution, compute the resulting GW transfer function, and obtain the full early-Universe GW spectrum in RDCC. We identify the frequency range in which relaxation-induced damping is observationally relevant, quantify the sectoral GW asymmetry, and provide concrete predictions for upcoming space-based interferometers. The results establish the SGWB as a sensitive probe of relaxation-driven early-Universe physics and extend the RDCC framework into the gravitational wave domain.

1 Motivation

The stochastic gravitational-wave background (SGWB) generated in the early Universe is a powerful probe of high-energy physics, providing access to epochs far beyond the reach of electromagnetic observations. In standard cosmology, the SGWB is determined by the primordial tensor spectrum and its subsequent evolution through the radiation- and matter-dominated eras. The resulting spectrum is smooth, nearly scale-invariant, and free of dynamical features except for those associated with changes in the relativistic degrees of freedom.

In the Relaxation-Driven Cyclic Cosmology (RDCC), the situation is fundamentally different. The universal relaxation mechanism introduced in Companion X modifies the evolution of super-horizon tensor modes during the earliest stages of cosmic history. When the relaxation rate satisfies $\Gamma(a) \gtrsim H(a)$, tensor perturbations experience an additional damping term that

suppresses their amplitude prior to horizon re-entry. This effect leaves a distinct imprint on the SGWB, producing a characteristic suppression feature at frequencies corresponding to modes that re-entered the horizon during the relaxation-dominated epoch.

A second key ingredient is the sectoral temperature asymmetry between the two CPT-conjugate sectors. As shown in Companion X, the temperature ratio $T_B/T_A \simeq 0.5$ implies different horizon masses, expansion histories, and damping scales in the two sectors. Tensor modes in the colder sector experience a different relaxation history, leading to a sectoral asymmetry in the GW spectrum. While only the visible-sector spectrum is directly observable, the mirror-sector spectrum influences the global energy balance and provides an internal consistency check of the RDCC framework.

The purpose of this Companion is to develop the early-Universe gravitational-wave sector of RDCC. We derive the relaxation-corrected tensor-mode equation, compute the resulting transfer function, and obtain the full SGWB spectrum including the relaxation-induced suppression feature. We identify the frequency range in which the effect is observationally relevant and provide concrete predictions for upcoming space-based interferometers such as LISA, DECIGO, and BBO.

Understanding the GW spectrum in RDCC is essential for three reasons:

- **Early-Universe dynamics:** Tensor modes directly probe the relaxation-dominated epoch, providing a clean test of the universal relaxation mechanism.
- **Sectoral structure:** The temperature asymmetry between the two sectors leads to a sectoral GW asymmetry, offering an independent probe of the bipartite global state.
- **Observational leverage:** The relaxation-induced suppression feature lies in the frequency range accessible to next-generation GW detectors, enabling direct observational tests of RDCC.

This Companion establishes the SGWB as a sensitive probe of relaxation-driven early-Universe physics and extends the RDCC framework into the gravitational-wave domain.

2 Tensor Modes in Standard Cosmology (Kurzüberblick)

In standard cosmology, primordial tensor perturbations originate from quantum fluctuations during inflation and evolve linearly throughout the early Universe. Their dynamics are governed by the tensor-mode equation

$$\ddot{h}_k + 3H\dot{h}_k + \frac{k^2}{a^2}h_k = 0, \quad (1)$$

where h_k denotes the Fourier amplitude of the transverse-traceless metric perturbation. The evolution naturally separates into two regimes:

- **Super-horizon regime** ($k \ll aH$): Tensor modes freeze out and maintain an approximately constant amplitude.
- **Sub-horizon regime** ($k \gg aH$): Tensor modes oscillate with frequency k/a and their amplitude decays as a^{-1} due to Hubble friction.

The primordial tensor power spectrum is typically parameterized as

$$\mathcal{P}_h(k) = A_t \left(\frac{k}{k_*} \right)^{n_t}, \quad (2)$$

with A_t the tensor amplitude and n_t the tensor tilt. In standard cosmology, the subsequent evolution of the stochastic gravitational-wave background (SGWB) is determined by the transfer

function $T(k)$, which encodes the effects of horizon re-entry, changes in the relativistic degrees of freedom, and the transition from radiation to matter domination.

The present-day energy density in gravitational waves is given by

$$\Omega_{\text{GW}}(k) = \frac{1}{12} \left(\frac{k}{a_0 H_0} \right)^2 \mathcal{P}_h(k) T^2(k), \quad (3)$$

where a_0 and H_0 denote the present-day scale factor and Hubble parameter. In the standard picture, the transfer function is smooth and featureless except for small steps associated with changes in g_* .

Crucially, the standard tensor-mode evolution contains *no additional damping terms* beyond Hubble friction. As a result, the SGWB spectrum is free of dynamical suppression features and retains a nearly scale-invariant shape across many decades in frequency.

This simplicity makes the SGWB an exceptionally sensitive probe of any new early-Universe physics that modifies tensor-mode evolution. In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduces precisely such a modification, as we develop in the next section.

3 Tensor Modes in RDCC: Relaxation-Induced Damping

In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduced in Companion X modifies the evolution of tensor perturbations during the earliest stages of cosmic history. The relaxation rate $\Gamma(a)$ governs the flow of information between the two CPT-conjugate sectors and introduces an additional damping term in the tensor-mode equation of motion. This effect is absent in standard cosmology and leads to a distinctive suppression feature in the stochastic gravitational-wave background (SGWB).

3.1 Tensor-Mode Equation with Relaxation

The evolution of a tensor mode h_k in RDCC is governed by the modified equation

$$\ddot{h}_k + (3H + \Gamma) \dot{h}_k + \frac{k^2}{a^2} h_k = 0, \quad (4)$$

where the additional term $\Gamma \dot{h}_k$ represents relaxation-induced damping. The relaxation rate satisfies

$$\Gamma(a) = \alpha(a) H(a), \quad (5)$$

with $\alpha(a)$ a slowly varying dimensionless coefficient of order $\mathcal{O}(0.1 - 1)$ during the relaxation-dominated epoch.

When $\Gamma \gg H$, the damping term dominates and suppresses the amplitude of tensor modes even while they remain super-horizon. This is in stark contrast to standard cosmology, where super-horizon tensor modes remain constant.

3.2 Super-Horizon Evolution

In the regime $k \ll aH$, the tensor-mode equation reduces to

$$\ddot{h}_k + (3 + \alpha) H \dot{h}_k \simeq 0. \quad (6)$$

Assuming $H \propto a^{-2}$ during radiation domination, the solution takes the form

$$h_k(a) \propto a^{-\alpha/2}. \quad (7)$$

Thus, super-horizon tensor modes are suppressed by a factor

$$S(a) = \left(\frac{a}{a_{\text{in}}}\right)^{-\alpha/2}, \quad (8)$$

where a_{in} denotes the scale factor at the onset of the relaxation-dominated epoch. This suppression persists until $\Gamma(a)$ drops below $H(a)$.

3.3 Sub-Horizon Evolution

Once a mode re-enters the horizon ($k \gg aH$), the equation becomes

$$\ddot{h}_k + (3H + \Gamma) \dot{h}_k + \frac{k^2}{a^2} h_k = 0. \quad (9)$$

In this regime, the mode oscillates with frequency k/a , while its amplitude decays as

$$h_k(a) \propto a^{-1} \exp\left[-\int \frac{\Gamma}{2H} \frac{da}{a}\right]. \quad (10)$$

The exponential factor represents the cumulative effect of relaxation-induced damping during and after horizon re-entry.

3.4 Physical Interpretation

The relaxation mechanism affects tensor modes in two ways:

- **Super-horizon suppression:** Tensor modes lose amplitude before horizon re-entry, reducing their contribution to the SGWB at the corresponding frequencies.
- **Enhanced sub-horizon damping:** The additional friction term $\Gamma \dot{h}_k$ accelerates the decay of tensor modes after horizon re-entry.

Both effects lead to a characteristic suppression feature in the SGWB spectrum, with the suppression scale determined by the epoch in which $\Gamma(a) \sim H(a)$.

3.5 Sectoral Asymmetry

Because the two CPT-conjugate sectors have different temperatures, their relaxation histories differ. The colder sector B has a smaller relaxation coefficient,

$$\alpha_B(a) < \alpha_A(a), \quad (11)$$

leading to weaker damping and a larger tensor amplitude. This results in a sectoral GW asymmetry, providing an independent probe of the bipartite global state.

In the next section, we compute the relaxation-corrected transfer function and derive the resulting SGWB spectrum.

4 Relaxation-Corrected Transfer Function

The transfer function $T(k)$ encodes the evolution of tensor modes from their primordial generation to the present day. In standard cosmology, $T(k)$ is determined by horizon re-entry, changes in the relativistic degrees of freedom, and the transition from radiation to matter domination. In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduces an additional suppression factor that modifies the transfer function on scales that re-entered the horizon during the relaxation-dominated epoch.

4.1 Definition of the Transfer Function

The present-day tensor power spectrum is given by

$$\mathcal{P}_h(k, a_0) = \mathcal{P}_h(k, a_{\text{prim}}) T^2(k), \quad (12)$$

where $\mathcal{P}_h(k, a_{\text{prim}})$ is the primordial tensor spectrum and $T(k)$ encodes all subsequent evolution. In RDCC, the transfer function factorizes as

$$T(k) = T_{\text{std}}(k) T_{\text{rel}}(k), \quad (13)$$

where $T_{\text{std}}(k)$ is the standard transfer function and $T_{\text{rel}}(k)$ captures the relaxation-induced suppression.

4.2 Relaxation-Induced Suppression Factor

From the tensor-mode evolution derived in Section 3, the amplitude of a tensor mode is suppressed by two effects:

- super-horizon suppression by the factor $S(a)$,
- enhanced sub-horizon damping due to the additional friction term $\Gamma \dot{h}_k$.

The combined suppression factor can be written as

$$T_{\text{rel}}(k) = S(a_{\text{re}}) \exp \left[- \int_{a_{\text{re}}}^{a_{\text{end}}} \frac{\Gamma(a)}{2H(a)} \frac{da}{a} \right], \quad (14)$$

where a_{re} is the scale factor at horizon re-entry and a_{end} marks the end of the relaxation-dominated epoch.

Using $\Gamma(a) = \alpha(a)H(a)$, this becomes

$$T_{\text{rel}}(k) = S(a_{\text{re}}) \exp \left[- \frac{1}{2} \int_{a_{\text{re}}}^{a_{\text{end}}} \alpha(a) \frac{da}{a} \right]. \quad (15)$$

If $\alpha(a)$ varies slowly, the integral simplifies to

$$T_{\text{rel}}(k) \simeq S(a_{\text{re}}) \left(\frac{a_{\text{re}}}{a_{\text{end}}} \right)^{\alpha/2}. \quad (16)$$

4.3 Scale Dependence

The relaxation-induced suppression affects only modes that re-entered the horizon while $\Gamma(a) \gtrsim H(a)$. Defining the characteristic scale k_{rel} by

$$k_{\text{rel}} = a_{\text{rel}} H(a_{\text{rel}}), \quad (17)$$

where a_{rel} satisfies $\Gamma(a_{\text{rel}}) = H(a_{\text{rel}})$, we obtain:

- $k \ll k_{\text{rel}}$: Modes re-enter after relaxation ends. No suppression: $T_{\text{rel}}(k) \simeq 1$.
- $k \gg k_{\text{rel}}$: Modes re-enter during relaxation. Strong suppression: $T_{\text{rel}}(k) \propto k^{-\alpha/2}$.

Thus, the relaxation mechanism introduces a *power-law suppression feature* in the SGWB spectrum at high frequencies.

4.4 Sectoral Asymmetry

Because the two CPT-conjugate sectors have different temperatures, their relaxation coefficients differ:

$$\alpha_B(a) < \alpha_A(a). \quad (18)$$

This implies

$$T_{\text{rel}}^{(B)}(k) > T_{\text{rel}}^{(A)}(k), \quad (19)$$

leading to a sectoral GW asymmetry. While only the visible-sector spectrum is directly observable, the mirror-sector spectrum provides an internal consistency check of the RDCC framework.

4.5 Resulting Transfer Function

Combining the standard and relaxation-induced contributions yields

$$T(k) = T_{\text{std}}(k) S(a_{\text{re}}) \exp\left[-\frac{1}{2} \int_{a_{\text{re}}}^{a_{\text{end}}} \alpha(a) \frac{da}{a}\right]. \quad (20)$$

This transfer function forms the basis for computing the full stochastic gravitational-wave background in RDCC, which we derive in the next section.

5 The Early-Universe GW Spectrum in RDCC

With the relaxation-corrected transfer function derived in Section 4, we now compute the full stochastic gravitational-wave background (SGWB) in the Relaxation-Driven Cyclic Cosmology (RDCC). The resulting spectrum exhibits a characteristic suppression feature at high frequencies, reflecting the damping of tensor modes during the relaxation-dominated epoch. This feature is absent in standard cosmology and provides a direct observational probe of relaxation-driven early-Universe physics.

5.1 General Expression

The present-day GW energy density per logarithmic interval in wavenumber is given by

$$\Omega_{\text{GW}}(k) = \frac{1}{12} \left(\frac{k}{a_0 H_0}\right)^2 \mathcal{P}_h(k) T^2(k), \quad (21)$$

where $\mathcal{P}_h(k)$ is the primordial tensor spectrum and $T(k)$ is the full transfer function,

$$T(k) = T_{\text{std}}(k) T_{\text{rel}}(k). \quad (22)$$

The primordial spectrum is parameterized as

$$\mathcal{P}_h(k) = A_t \left(\frac{k}{k_*}\right)^{n_t}, \quad (23)$$

with A_t the tensor amplitude and n_t the tensor tilt.

5.2 Relaxation-Induced Suppression Feature

Using the relaxation-corrected transfer function from Section 4, the GW spectrum becomes

$$\Omega_{\text{GW}}(k) = \Omega_{\text{GW}}^{\text{std}}(k) T_{\text{rel}}^2(k), \quad (24)$$

where $\Omega_{\text{GW}}^{\text{std}}(k)$ is the standard spectrum and

$$T_{\text{rel}}(k) \simeq \begin{cases} 1, & k \ll k_{\text{rel}}, \\ \left(\frac{k}{k_{\text{rel}}}\right)^{-\alpha/2}, & k \gg k_{\text{rel}}. \end{cases} \quad (25)$$

Thus, the relaxation mechanism introduces a power-law suppression at high frequencies:

$$\Omega_{\text{GW}}(k) \propto k^{n_t - \alpha}. \quad (26)$$

The break scale k_{rel} corresponds to the mode that re-entered the horizon when $\Gamma(a) = H(a)$.

5.3 Frequency Range of the Suppression Feature

The characteristic frequency associated with k_{rel} is

$$f_{\text{rel}} = \frac{k_{\text{rel}}}{2\pi a_0} = \frac{a_{\text{rel}} H(a_{\text{rel}})}{2\pi a_0}. \quad (27)$$

For typical RDCC parameters, the relaxation-dominated epoch occurs at temperatures

$$T \sim 10^7 - 10^{10} \text{ GeV}, \quad (28)$$

corresponding to frequencies in the range

$$f_{\text{rel}} \sim 10^{-3} - 10^2 \text{ Hz}. \quad (29)$$

This range overlaps with the sensitivity windows of LISA, DECIGO, and BBO, making the relaxation-induced suppression feature directly observable.

5.4 Sectoral GW Asymmetry

Because the two CPT-conjugate sectors have different temperatures, their relaxation coefficients differ:

$$\alpha_B < \alpha_A. \quad (30)$$

This implies

$$\Omega_{\text{GW}}^{(B)}(k) > \Omega_{\text{GW}}^{(A)}(k) \quad (31)$$

for $k \gg k_{\text{rel}}$, leading to a sectoral GW asymmetry.

While only the visible-sector spectrum is directly observable, the mirror-sector spectrum affects the global energy balance and provides an internal consistency check of the RDCC framework.

5.5 Full RDCC GW Spectrum

Combining all contributions, the RDCC GW spectrum takes the form

$$\Omega_{\text{GW}}(k) = \frac{1}{12} \left(\frac{k}{a_0 H_0}\right)^2 A_t \left(\frac{k}{k_*}\right)^{n_t} T_{\text{std}}^2(k) T_{\text{rel}}^2(k), \quad (32)$$

with

$$T_{\text{rel}}(k) \simeq \begin{cases} 1, & k \ll k_{\text{rel}}, \\ \left(\frac{k}{k_{\text{rel}}}\right)^{-\alpha/2}, & k \gg k_{\text{rel}}. \end{cases} \quad (33)$$

This spectrum exhibits a smooth, nearly scale-invariant low-frequency regime and a relaxation-induced suppression at high frequencies. The location and slope of the suppression feature provide direct information about the relaxation dynamics and the temperature asymmetry between the two sectors.

In the next section, we translate these results into concrete observational predictions for upcoming gravitational-wave detectors.

6 Observational Predictions

The relaxation-corrected gravitational-wave (GW) spectrum derived in Section 5 leads to a set of concrete and testable predictions. These predictions arise from the interplay between the relaxation-induced damping of tensor modes, the sectoral temperature asymmetry, and the frequency dependence of horizon re-entry. In this section we summarize the most important observational consequences for space-based interferometers and high-frequency GW searches.

Prediction 1: High-Frequency Suppression Feature

The universal relaxation mechanism introduces a characteristic suppression feature in the stochastic gravitational-wave background (SGWB) at frequencies

$$f \gtrsim f_{\text{rel}} \sim 10^{-3} - 10^2 \text{ Hz}. \quad (34)$$

Modes that re-entered the horizon during the relaxation-dominated epoch experience a power-law suppression,

$$\Omega_{\text{GW}}(f) \propto f^{n_t - \alpha}, \quad (35)$$

with α the relaxation coefficient. This feature is absent in standard cosmology and provides a direct probe of relaxation-driven early-Universe physics.

Prediction 2: Observable Break Scale

The break frequency f_{rel} marks the transition between the unsuppressed and suppressed regimes. For typical RDCC parameters, f_{rel} lies within the sensitivity windows of LISA, DECIGO, and BBO. A detection of a spectral break at this frequency would provide strong evidence for the relaxation mechanism.

Prediction 3: Modified Spectral Slope

Above the break scale, the GW spectrum exhibits a modified slope,

$$\frac{d \ln \Omega_{\text{GW}}}{d \ln f} = n_t - \alpha, \quad (36)$$

where α is of order $\mathcal{O}(0.1 - 1)$ during the relaxation epoch. This leads to a significant deviation from the nearly scale-invariant spectrum predicted by standard cosmology.

Prediction 4: Sectoral GW Asymmetry

Because the two CPT-conjugate sectors have different temperatures, their relaxation coefficients differ:

$$\alpha_B < \alpha_A. \quad (37)$$

This implies a sectoral asymmetry in the GW spectrum,

$$\Omega_{\text{GW}}^{(B)}(f) > \Omega_{\text{GW}}^{(A)}(f) \quad (38)$$

for $f \gtrsim f_{\text{rel}}$. While only the visible-sector spectrum is directly observable, the mirror-sector spectrum affects the global energy budget and provides an internal consistency check of the RDCC framework.

Prediction 5: Consistency with PBH Constraints

The relaxation-induced suppression of tensor modes is consistent with the suppression of primordial black hole (PBH) formation derived in Companion XIV. Both effects originate from the same relaxation dynamics and occur at similar energy scales. A joint analysis of PBH constraints and GW observations can therefore test the internal coherence of RDCC.

Prediction 6: Distinguishability from Other New-Physics Scenarios

The relaxation-induced suppression feature differs qualitatively from other mechanisms that modify the SGWB, such as:

- changes in relativistic degrees of freedom (g_*),
- phase transitions,
- cosmic strings,
- reheating effects.

These scenarios typically produce sharp features, bumps, or oscillations, whereas RDCC predicts a smooth power-law suppression with a well-defined break scale. This distinct spectral shape allows the relaxation mechanism to be distinguished from other early-Universe physics.

Prediction 7: Direct Testability with Future Detectors

The predicted suppression feature lies in the frequency range accessible to upcoming space-based interferometers. In particular:

- **LISA** can probe the low-frequency edge of the suppression feature.
- **DECIGO** and **BBO** can probe the full suppression regime.
- **High-frequency detectors** (MHz–GHz) can test the extreme tail of the spectrum.

A detection of the predicted spectral break and modified slope would provide strong evidence for the relaxation mechanism and the bipartite structure of RDCC.

Together, these predictions establish the SGWB as a sensitive probe of relaxation-driven early-Universe physics and provide multiple observational avenues for testing the RDCC framework.

7 Conclusion

In this Companion we have developed the gravitational-wave (GW) sector of the Relaxation-Driven Cyclic Cosmology (RDCC), extending the framework into a domain that is directly accessible to upcoming space-based interferometers. Starting from the universal relaxation mechanism introduced in Companion X, we derived the relaxation-corrected evolution of tensor perturbations, computed the corresponding transfer function, and obtained the full early-Universe stochastic gravitational-wave background (SGWB) in RDCC.

The key result is that the relaxation mechanism introduces a characteristic suppression feature in the GW spectrum at frequencies corresponding to modes that re-entered the horizon during the relaxation-dominated epoch. This suppression arises from two effects: the relaxation-induced damping of super-horizon tensor modes and the enhanced friction term affecting sub-horizon evolution. The resulting spectrum exhibits a smooth power-law suppression above a

well-defined break scale, providing a clear observational signature of relaxation-driven early-Universe physics.

A second important result is the sectoral GW asymmetry. Because the two CPT-conjugate sectors have different temperatures, their relaxation histories differ, leading to distinct GW spectra. While only the visible-sector spectrum is directly observable, the mirror-sector spectrum contributes to the global energy balance and provides an internal consistency check of the RDCC framework. This sectoral asymmetry parallels the sectoral PBH asymmetry developed in Companion XIV, demonstrating the coherence of the relaxation mechanism across multiple cosmological observables.

The predicted suppression feature lies within the sensitivity windows of LISA, DECIGO, and BBO, making the SGWB a powerful probe of RDCC. A detection of the spectral break and modified slope would provide strong evidence for the relaxation mechanism and the bipartite structure of the early Universe. Conversely, the absence of such a feature would place meaningful constraints on the relaxation coefficient and the duration of the relaxation-dominated epoch.

With the development of the GW sector, RDCC now provides a coherent and internally consistent description of scalar, tensor, and PBH-related early-Universe physics. The framework continues to yield a rich set of observational signatures, positioning the SGWB as a central tool for testing the relaxation-driven dynamics that define the RDCC cosmology.

Appendix A: Derivation of the Relaxation-Corrected Tensor Equation

In this appendix we derive the relaxation-corrected evolution equation for tensor perturbations used in the main text. The key modification relative to standard cosmology arises from the universal relaxation mechanism introduced in Companion X, which governs the flow of information between the two CPT-conjugate sectors and introduces an additional damping term in the tensor-mode dynamics.

A.1 Tensor Modes in Standard Cosmology

We start from the perturbed Friedmann–Lemaître–Robertson–Walker (FLRW) metric with a transverse-traceless tensor perturbation h_{ij} ,

$$ds^2 = -dt^2 + a^2(t) (\delta_{ij} + h_{ij}) dx^i dx^j, \quad (39)$$

with

$$\partial_i h^{ij} = 0, \quad h^i_i = 0. \quad (40)$$

In standard cosmology, the linearized Einstein equations yield the tensor-mode equation

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{1}{a^2}\nabla^2 h_{ij} = 0. \quad (41)$$

In Fourier space, this becomes

$$\ddot{h}_k + 3H\dot{h}_k + \frac{k^2}{a^2}h_k = 0, \quad (42)$$

where h_k denotes the Fourier amplitude of a given polarization mode.

A.2 Inclusion of the Relaxation Mechanism

In the Relaxation-Driven Cyclic Cosmology (RDCC), the universal relaxation mechanism introduces an additional channel for energy and information exchange between the two sectors.

At the level of the effective description, this appears as an extra damping term in the equation of motion for tensor modes. The modified equation takes the form

$$\ddot{h}_k + (3H + \Gamma) \dot{h}_k + \frac{k^2}{a^2} h_k = 0, \quad (43)$$

where $\Gamma(a)$ is the relaxation rate.

The relaxation rate is parametrized as

$$\Gamma(a) = \alpha(a) H(a), \quad (44)$$

with $\alpha(a)$ a slowly varying dimensionless coefficient. During the relaxation-dominated epoch, $\alpha(a)$ is of order $\mathcal{O}(0.1 - 1)$, implying $\Gamma \gtrsim H$.

A.3 Super-Horizon Limit

In the super-horizon regime, $k \ll aH$, the gradient term can be neglected and the equation reduces to

$$\ddot{h}_k + (3 + \alpha) H \dot{h}_k \simeq 0. \quad (45)$$

During radiation domination, $H \propto a^{-2}$, and we can write

$$\frac{d}{dt} = aH \frac{d}{da}. \quad (46)$$

Using this, the equation becomes

$$aH \frac{d}{da} \left(aH \frac{dh_k}{da} \right) + (3 + \alpha) H \left(aH \frac{dh_k}{da} \right) \simeq 0. \quad (47)$$

Neglecting the slow variation of α and using $H \propto a^{-2}$, one finds the approximate solution

$$h_k(a) \propto a^{-\alpha/2}. \quad (48)$$

Thus, super-horizon tensor modes are suppressed by the factor

$$S(a) = \left(\frac{a}{a_{\text{in}}} \right)^{-\alpha/2}, \quad (49)$$

where a_{in} denotes the scale factor at the onset of the relaxation-dominated epoch.

A.4 Sub-Horizon Limit

In the sub-horizon regime, $k \gg aH$, the tensor-mode equation becomes

$$\ddot{h}_k + (3H + \Gamma) \dot{h}_k + \frac{k^2}{a^2} h_k = 0. \quad (50)$$

For modes well inside the horizon, the solution can be written as

$$h_k(a) = A_k(a) \cos \left(\int \frac{k}{a} dt + \varphi_k \right), \quad (51)$$

where $A_k(a)$ is a slowly varying amplitude and φ_k a phase. Inserting this ansatz and averaging over oscillations yields an evolution equation for the envelope $A_k(a)$,

$$\frac{d \ln A_k}{d \ln a} = -1 - \frac{1}{2} \frac{\Gamma}{H}. \quad (52)$$

Integrating from horizon re-entry at a_{re} to the end of the relaxation-dominated epoch at a_{end} gives

$$A_k(a_{\text{end}}) = A_k(a_{\text{re}}) \left(\frac{a_{\text{re}}}{a_{\text{end}}} \right) \exp \left[-\frac{1}{2} \int_{a_{\text{re}}}^{a_{\text{end}}} \frac{\Gamma(a)}{H(a)} \frac{da}{a} \right]. \quad (53)$$

Using $\Gamma(a) = \alpha(a)H(a)$, this becomes

$$A_k(a_{\text{end}}) = A_k(a_{\text{re}}) \left(\frac{a_{\text{re}}}{a_{\text{end}}} \right) \exp \left[-\frac{1}{2} \int_{a_{\text{re}}}^{a_{\text{end}}} \alpha(a) \frac{da}{a} \right]. \quad (54)$$

A.5 Combined Suppression Factor

Combining the super-horizon suppression and the enhanced sub-horizon damping, the total suppression factor for a mode k can be written as

$$T_{\text{rel}}(k) = S(a_{\text{re}}) \exp \left[-\frac{1}{2} \int_{a_{\text{re}}}^{a_{\text{end}}} \alpha(a) \frac{da}{a} \right]. \quad (55)$$

If $\alpha(a)$ varies slowly, the integral simplifies to

$$T_{\text{rel}}(k) \simeq S(a_{\text{re}}) \left(\frac{a_{\text{re}}}{a_{\text{end}}} \right)^{\alpha/2}. \quad (56)$$

This expression is the basis for the relaxation-corrected transfer function used in the main text and encapsulates the impact of the universal relaxation mechanism on tensor-mode evolution in RDCC.

Appendix B: Sectoral Differences in the GW Spectrum

In this appendix we analyze how the temperature asymmetry between the two CPT-conjugate sectors of the Relaxation-Driven Cyclic Cosmology (RDCC) affects the gravitational-wave (GW) spectrum. As shown in Companion X, the temperature ratio

$$\frac{T_B}{T_A} \simeq 0.5 \quad (57)$$

implies that sector B is colder throughout the early Universe. This temperature difference modifies the relaxation coefficient, the horizon re-entry scale, and the damping history of tensor modes in each sector, leading to a sectoral asymmetry in the GW spectrum.

B.1 Temperature Dependence of the Relaxation Coefficient

The relaxation rate satisfies

$$\Gamma(a) = \alpha(a) H(a), \quad (58)$$

with $\alpha(a)$ a slowly varying dimensionless coefficient. Because the relaxation mechanism is thermodynamic in origin, the colder sector exhibits a smaller relaxation coefficient:

$$\alpha_B(a) < \alpha_A(a). \quad (59)$$

This implies weaker relaxation-induced damping in sector B.

B.2 Sectoral Horizon Re-Entry Scales

The characteristic scale at which relaxation ceases to dominate is defined by

$$k_{\text{rel}}^{(i)} = a_{\text{rel}}^{(i)} H(a_{\text{rel}}^{(i)}), \quad (60)$$

where $i \in \{A, B\}$ labels the sector. Because the colder sector expands more slowly at fixed scale factor, we have

$$k_{\text{rel}}^{(B)} < k_{\text{rel}}^{(A)}. \quad (61)$$

Thus, the relaxation-induced suppression feature begins at a lower frequency in sector B.

B.3 Sectoral Transfer Functions

The relaxation-corrected transfer function in each sector is

$$T_{\text{rel}}^{(i)}(k) = S^{(i)}(a_{\text{re}}^{(i)}) \exp \left[-\frac{1}{2} \int_{a_{\text{re}}^{(i)}}^{a_{\text{end}}^{(i)}} \alpha_i(a) \frac{da}{a} \right]. \quad (62)$$

Because $\alpha_B < \alpha_A$, we obtain

$$T_{\text{rel}}^{(B)}(k) > T_{\text{rel}}^{(A)}(k) \quad (63)$$

for all modes that re-enter during the relaxation-dominated epoch.

B.4 Sectoral GW Spectra

The GW energy density in each sector is

$$\Omega_{\text{GW}}^{(i)}(k) = \Omega_{\text{GW}}^{\text{std}}(k) \left[T_{\text{rel}}^{(i)}(k) \right]^2. \quad (64)$$

For $k \gg k_{\text{rel}}^{(i)}$, the spectrum exhibits a power-law suppression,

$$\Omega_{\text{GW}}^{(i)}(k) \propto k^{n_t - \alpha_i}. \quad (65)$$

Since $\alpha_B < \alpha_A$, the colder sector has a shallower suppression slope:

$$\Omega_{\text{GW}}^{(B)}(k) > \Omega_{\text{GW}}^{(A)}(k) \quad (66)$$

at high frequencies.

B.5 Physical Interpretation

The sectoral GW asymmetry arises from three effects:

- **Different relaxation strengths:** The colder sector experiences weaker damping.
- **Different re-entry scales:** Modes re-enter earlier in the hotter sector, enhancing suppression.
- **Different super-horizon histories:** The suppression factor $S(a)$ differs between sectors.

These effects combine to produce a robust sectoral asymmetry in the GW spectrum, analogous to the sectoral PBH asymmetry derived in Companion XIV.

B.6 Observational Implications

While only the visible-sector spectrum is directly observable, the mirror-sector spectrum affects the global energy budget and provides an internal consistency check of the RDCC framework. A joint analysis of the GW spectrum, PBH constraints, and sectoral thermodynamics can therefore test the coherence of the relaxation mechanism across multiple cosmological observables.

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